LUMINOSITIES

Beta Canum Venaticorum
1.2 Suns

RS Canum Venaticorum
13 Suns

CANES VENATICI
THE HUNTING DOGS

BETWEEN BOÖTES AND URSA MAJOR LIES THE CONSTELLATION CANES VENATICI, REPRESENTING A PAIR OF HUNTING DOGS HELD ON A LEASH BY BOÖTES. SEVERAL REMARKABLE GALAXIES LIE WITHIN ITS BORDERS, MOST NOTABLY M51, POPULARLY KNOWN AS THE WHIRLPOOL.

Not recognized by the ancient Greeks, the constellation Canes Venatici was introduced in 1687 by Johannes Hevelius, a Polish astronomer who invented several new sky figures. He imagined it as two hounds held on a leash by the adjacent Boötes, the Herdsman.

This constellation has few stars of note. Its brightest star was named Cor Caroli (Charles's Heart) in the 17th century to commemorate King Charles I of England, who was beheaded by the republican parliament in 1649.

Near the constellation's upper border with Ursa Major lies M51 (see pp.114–15), a face-on spiral galaxy. Its spiral structure was first detected in 1845 by an Irish astronomer, Lord Rosse, using a telescope he had built himself at his home at Birr Castle, County Offaly. Rosse's discovery led to speculation that such spiral objects could be separate galaxies far off in space. In the case of M51, the distance is about 30 million light-years. A smaller galaxy, called NGC 5195, lies near the end of one of its arms.

> In 1845, Lord Rosse observed M51, the **first spiral galaxy** to be recognized, using what was then the **world's largest telescope**

◁ **M106**
This view of spiral galaxy M106 is a composite of images from the Hubble Space Telescope and two amateur astrophotographers, Robert Gendler and Jay GaBany.

▽ **NGC 4449**
The glowing patches in this dwarf galaxy are bursts of star formation, most probably triggered by an interaction or merger with one or more smaller galaxies.

KEY DATA

Size ranking 38
Brightest stars Alpha (α) 2.9, Beta (β) 4.3
Genitive Canum Venaticorum
Abbreviation CVn
Highest in sky at 10pm April–May
Fully visible 90°N–27°S

CHART 5

MAIN STARS

Cor Caroli Alpha (α) Canum Venaticorum
Blue-white main sequence
2.9 ⟷ 115 light-years

Beta (β) Canum Venaticorum
Yellow main sequence
4.3 ⟷ 28 light-years

La Superba Y Canum Venaticorum
Red giant variable
4.9–7.3 ⟷ 1,000 light-years

RS Canum Venaticorum
Eclipsing binary
7.9–9.1 ⟷ 520 light-years

DEEP-SKY OBJECTS

M3
Globular cluster

M51
Spiral galaxy; also known as the Whirlpool Galaxy

M63
Spiral galaxy; also known as the Sunflower Galaxy

M94
Spiral galaxy

M106
Spiral galaxy

NGC 4244
Edge-on spiral galaxy

NGC 4449
Irregular dwarf galaxy

NGC 4631
Edge-on spiral galaxy; also called the Whale Galaxy